PLAN — Moon reading light

Status: **v0.1 design complete and through verification rounds 1 and 2, nothing ordered** (2026-11-02). The build pack is in `deliverable/` (start with `00-START-HERE.pdf`). Next: Matt orders boards + parts, prints the coupons, builds and bench-tests (docs/handoffs/, docs/verification/VERIFY.md). Single source of truth for requirements and decisions; reasoning is kept so later revisions don't re-derive it.

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1. Requirements

Statuses follow the 3dprint convention: **GIVEN** (Matt said it), **ASSUMED** (Claude's reading; confirm), **RULE** (engineering rule this design adopts).

#	Status	Requirement
R1	GIVEN	Clip-on reading light: a pack clips onto the back of a book with a slip-on clip
R2	GIVEN	Pack holds the battery and electronics; enclosed mini LiPo, USB-C charging
R3	GIVEN	From the top of the pack a flexible, adjustable neck ~10 cm long
R4	GIVEN	At the end of the neck a hollow model of the moon, ~5 cm diameter, lit fully from inside by LEDs
R5	GIVEN	One main button: press turns the light on/cycles colours (answer 3: the button never turns it off)
R6	GIVEN	Hold the main button > 2 s: sets the current colour as the power-on default
R7	GIVEN	Slide on/off switch on the side is the only on/off (answer 3)
R8	GIVEN	Up and down brightness buttons above and below the main button
R9	GIVEN	Colours: soft white, blue-white, yellow/orange/natural white, soft orange night light, red, green, blue; research and add other night-light whites
R10	GIVEN	Moon model generated from NASA data (answer 5), Thingiverse surveyed
R11	GIVEN	LED design: CRI-90+ tunable white + RGB with a constant-current driver (answer 1); LEDs in the base of the moon at the top of the neck
R12	GIVEN	Neck: pre-made metal, plastic-coated bendable arm
R13	GIVEN	Battery ~1200 mAh (answer 4)
R14	GIVEN	PCB files ready for PCBWay (or another fab); print files ready for the printer (Bambu X2D); BOM + order list for Farnell or RS
R15	GIVEN	Use the 14MM47/pcb and 14MM47/3dprint processes; product files live in this repo (answer 6)
R16	GIVEN	Multiple verification runs against datasheets, with adversarial/specialist reviewers
R17	RULE	Slide switch off = battery isolated from everything except the charger: standby drain < 10 µA
R18	RULE	LiPo charged CC/CV 4.20 V at ≤ 0.5 C; reverse-battery protected; load sharing so charging terminates with the light on
R19	RULE	Flicker-free (PWM ≥ 20 kHz) and no PLA moon softening: LED-board temperature limited in firmware with a sensor
R20	RULE	Night modes contain no green or blue light (melanopic research); whites on the Planckian locus
R21	ASSUMED	Hand-assembly friendly parts (0603+, SOIC/TSSOP, no QFN/BGA) so all three assembly routes work
R22	ASSUMED	Pack ≤ ~45 × 80 × 20 mm, total mass ≤ ~100 g

2. Architecture

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 USB-C (power only, 5.1k CC) —PTC 1.1 A——TVS—┬—> MCP73831 charger 256 mA —> BAT —[Q3 rev-batt
P-FET]— LiPo 1200 mAh (PCM)
                                              └—> D3 S1J rectifier —┐
 BAT —[Q1 load-share P-FET, gate = VBUS, R5 10k]———————————————————┴—> VSYS_RAW —[Q2 P-FET,
slide switch]—> VSYS 3.3–4.2 V
 VSYS —> ATtiny3216 (buttons, I2C master, VDD measurement, PB2 → Q4)
 VSYS —[Q4 P-FET, MCU-switched]—> VLED (+ I2C pull-ups) — JST-SH 4-wire cable up the gooseneck
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   LED board in the moon base (Ø26 mm, 1.0 mm): TLC59116 16-ch constant-current sink, 25
mA/channel, 97 kHz PWM
   ├— 4 × 2700 K + 4 × 5000 K Bridgelux Thrive 2835 (CRI 97)      ├— 2 × Würth RGB (common
anode)
   └— LM75B: firmware derating + OS thermostat → TLC RESET (60/52 °C)   all anodes on VLED
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- **Main board** (pack, 35 × 62 mm, 2-layer 1.6 mm, all parts top): see `hardware/main-board/design.py` docstring.
- **LED board** (moon base, Ø26 mm, 2-layer 1.0 mm, white mask; LEDs top, driver/sensor bottom): `hardware/led-board/design.py`.
- **Optics**: a Lambertian source on the wall of a sphere lights its inside uniformly (integrating-sphere geometry). Model `docs/design/optics/sphere_illum.py`: with the LED plane 1–2.5 mm below a 30–36° opening, wall illuminance max/min ≈ 1.00–1.04 for the whites (≤ 1.15 for single RGB colours, round1 F8) and ~97 % of the LED flux reaches the wall. Raising the board above the opening gives ≥ 2:1.
- **Light output**: 4000 K full ≈ 40 lm out of the moon, **20–30 lux on a page at 25–30 cm**. That is a moon lamp / night-reading level, not a full reading light (round1 F6; v0.2 option in docs/verification/round1.md §5).
- **Firmware**: `firmware/` — portable core (`lib/core`) + hardware glue (`src/main.cpp`); modes from `tools/gen_modes.py` (CIE 1931 mixing to the Planckian locus).
- **Mechanics**: lithophane moon (PLA, NASA LROC albedo + LOLA relief → 0.9–2.6 mm wall, relief inside), bayonet-locked to a PETG base cup. A 0.6 mm PETG ring clamps the LED board, so the PLA never touches it. The cup screws onto the gooseneck's M10×1 end. Pack = front shell + back plate + slip-on cantilever clip (PETG).
- **Thermal**: three layers. Firmware derating 45/52/58 °C; a 200 mA cap only with a trusted sensor (else 110 mA); the LM75B hardware thermostat. Detail: docs/design/electronics.md §2.

3. Light modes (R9, R20)

Ten modes in button order (`firmware/include/modes_table.h`, generated): Warm 2700 K · Soft white 3000 K · Neutral reading 4000 K · Daylight 5000 K · Blue-white 6500 K · Candle 1900 K · Soft orange night · Red night · Green · Blue. Evidence and targets: `docs/research/light-modes.md` (melanopic EDI limits from Brown et al. 2022, red for dark adaptation, Planckian chromaticities). 16 brightness steps, geometric 0.4 %–100 %. Power-on default: Soft white 3000 K at level 10 until the user saves another. Green and Blue run at half current for supply headroom. Night modes go to level 15 (still < 1 lux at 30 cm).

4. Decision log

- 2026-11-02 — Product files in `reading_light`; process conventions from `14MM47/pcb` and `14MM47/3dprint` (Matt, answer 6).

- 2026-11-02 — LEDs in the moon base facing up (Matt's suggestion); confirmed by the optics model (uniform within ~1 %).
- 2026-11-02 — Slide switch is the only on/off; main button never turns off (Matt, answer 3).
- 2026-11-02 — Farnell UK chosen over RS (wider semiconductor range, Adafruit/TI/Microchip stock); RS only as fallback.
- 2026-11-02 — LED driver TLC59116 (Farnell 3119169) instead of TI LP5012 (not stocked by Farnell/RS). Cost: 25–37 mA driver supply current while on; 8-bit PWM (+ IREF gain for low-end resolution).
- 2026-11-02 — Cool white is 5000 K (only CRI-90+ cool 2835 at Farnell); 6500 K is mixed with blue + a little green.
- 2026-11-02 — RGB LED Würth 150141M173100 (datasheet + exact KiCad footprint) instead of Kingbright KAAF-5050 (datasheet unobtainable → pinout unverifiable). Würth abs max 30 mA → TLC59116 REXT 750 Ω = 25.0 mA/channel.
- 2026-11-02 — Two RGB LEDs (14 of 16 driver channels) to double red for candle/night modes.
- 2026-11-02 — USB→system OR-ing uses an S1J silicon rectifier, not a Schottky: MBR120VLSF reverse leakage (0.6 mA max @25 °C) would drain the cell through the load-share FET while switched off. Cost: VSYS ≈ 4.15 V on USB.
- 2026-11-02 — 10 µF capacitors are 0805 16 V (0603 10 V keeps only ~2.9 µF at 5 V per Samsung's DC-bias data).
- 2026-11-02 — LED board Ø26 mm (moon opening 33° instead of 30°): the 24 mm board could not be routed; optics model shows no uniformity change (max/min 1.001 at the chosen depth).
- 2026-11-02 — LM75BDP (TSSOP-8) instead of LM75BD (SOIC-8) to fit the LED board.
- 2026-11-02 — Moon relief on the inside, smooth 50.0 mm outside (crisp silhouette; outer overhangs ≤ 45° by claim).
- 2026-11-02 — Gooseneck: 120 mm, 10 mm OD, M10×1 ends, 6.8 mm bore (no coated gooseneck of this length is sold in the UK); sleeve it in silicone/heat-shrink for the plastic-coated look and feel (R12).
- 2026-11-02 — Verification round 1 (4 reviewers; docs/verification/round1.md). Changes adopted:
  - `board_map.h` is generated by `hardware/build.py`, with a consistency test (B1).
  - Q4 MCU-switched LED-board supply; pull-ups on VLED (M2, M-3).
  - J4 pin 1 = VREF via 1 k (M3).
  - Charge 256 mA, R3 3.9 k (M4).
  - F1 PTC ahead of the TVS (M-5, M-6).
  - R5 10 k (m1).
  - LM75B OS hardware thermostat on TLC RESET (M-1, m6).
  - Derating 45/52/58 °C, with the trusted-sensor 200/110 mA caps (F1, F4).
  - PETG clamp ring (F5).
  - Recipes regenerated for the Würth RGB at 25 mA (F7).
  - CG steps limited to the datasheet ranges (F9).
  - Green/Blue at half current (M1).
  - Firmware: WDT, bus recovery, read-back, timer-wrap fixes.
  - Pack: J2 relocated with a lead slot (B-3); 1.6 mm screw-head gap over the pouch (B-2).
  - LED board 1.0 mm in the board file (N9).
  - Router fixed to keep the whole track body out of rule areas.
- 2026-11-02 — Verification round 2 (2 reviewers; docs/verification/round2.md). Changes adopted:
  - Battery-lead slot 6.4 × 5.0 mm, so the pre-fitted plug passes.

    - Q4 soft turn-on (R13 10 k, C8 100 nF gate–drain).
    - F1 is now MF-MSMF110/16X; TP7 VLED test pad added.
    - LED-board RESET pull-up 2.4 k.
    - W/C silk letters at the white LEDs; values and DNP cross-outs on the assembly drawings.
    - Firmware: `cg_select()` moved into the host-tested core; faded cap changes; frozen timer from load start; SWRST and VLED bleed on power-cycle; LM75B read-back.
    - CAD: back-plate locators inside the cavity; 4.5 mm under the LED board; 0.88 mm board pocket; J2 headroom pocket; switch finger recess; larger USB-C window; button flange Ø9.0.
    - New `jig_led_board`; cross-part claims added.
- 2026-11-02 — `fab/*/v0.1/` committed before ordering (deviation from the pcb repo's "commit fab on order" rule): Matt asked for files he can take straight to the fab. The order reference and tag follow at order time (handoff 01).
- 2026-11-02 — Accepted for v0.1, revisit in v0.2: no cell NTC; reversed cell not blocked while USB is present; G/B tint drift at low battery; reading illuminance 20–30 lux. Decisions for Matt are listed in round1.md §5.

5. Open questions / VERIFY

See `docs/verification/VERIFY.md` (measurements that only the prototype can settle).

6. Status

Phase	Deliverable	Status
Requirements & research	this file, docs/research	done
Electronics	hardware/*, ERC 0 / DRC 0 errors / 0 unconnected / 0 parity; fab/*v0.1	done (v0.1)
Firmware	firmware/, 29 host tests + AVR builds	done (v0.1), untested on hardware
Mechanics	cad/parts (11 parts PASS incl. the LED-board jig), print/ STL + 3MF	done (v0.1)
Verification rounds	docs/verification/round1.md, round2.md, VERIFY.md	rounds 1 and 2 done, all fixes applied and rebuilt
BOM/order, handoffs	bom/, docs/handoffs 01–06	done
Build pack	deliverable/ (generated by tools/build_deliverable.py)	done